VyapaarSetu

AI-Powered MSE Agent Mapping Tool for ONDC TEAM Initiative

Solution Deployment Details

Current Deployment Status: Development / Proof of Concept (PoC) Stage

Problem Statement:	AI-Powered MSE Agent Mapping Tool (PS2)
Ministry:	Ministry of Micro, Small and Medium Enterprises (MSME)
Challenge:	IndiaAI Innovation Challenge 2026
Document Version:	1.0
Date:	February 2026

1. Deployment Architecture

VyapaarSetu is deployed as a containerized multi-service architecture, enabling single-command deployment and horizontal scalability. The solution follows microservices best practices with clear separation of concerns across components.

Component	Technology	Port	Purpose
Backend API	FastAPI (Python)	8000	Core business logic, LangChain agent orchestration, skill-based AI pipeline
Frontend	Next.js 16	3000	Three portals — MSE onboarding, SNP lead management, NSIC admin dashboard
WhatsApp Service	Baileys (Node.js)	3001	WhatsApp integration for catalogue building and milestone notifications
Database	PostgreSQL	5432	Persistent storage for MSEs, SNPs, products, matches, sessions
Cache Layer	Redis	6379	Session management, rate limiting, conversation state
Vector Database	Qdrant	6333	RAG knowledge base with ONDC taxonomy and TEAM scheme embeddings

Containerization: All services are containerized via Docker Compose, enabling single-command deployment with `docker-compose up -d`. This ensures consistent environments across development, testing, and production deployments.

2. Current Deployment Environment

Parameter	Current Value
Environment Type	Local Development Server
Host Operating System	Windows 11
Container Runtime	Docker Desktop with WSL2
Data Seeding	Demo MSEs, 6 real ONDC SNPs, 10 ONDC RET categories
RAG Knowledge Base	Indexed ONDC taxonomy, TEAM scheme guidelines, SNP capability profiles

Demo User Configuration:

- Owner: Rajesh Kumar
- Enterprise: Kumar Handicrafts
- UDYAM Registration: UDYAM-MH-01-0012345
- Mobile: Registered for identity verification testing
- Products: Brass diyas, handicraft items (ONDC:RET16 — Home & Decor)

3. Testing Coverage

The following functionalities have been tested in the current PoC deployment:

Category	Feature Tested	Status
Voice Interface	Voice-based product catalogue building (Hindi/Hinglish) via web browser	Tested
Voice Interface	Multi-turn conversational field collection (7 product attributes)	Tested
Identity	Udyam number verification and auto-fill of business profile	Tested
Cataloguing	AI product classification to ONDC RET taxonomy (10 categories)	Tested
Cataloguing	NER-based attribute extraction from natural language descriptions	Tested
Compliance	E-Commerce Rules 2020 + Legal Metrology Act validation scoring	Tested
Messaging	WhatsApp message dispatch and reply handling via Baileys	Tested
Matching	MSE-to-SNP intelligent matching (6 real ONDC SNPs)	Tested
AI Agent	ReAct agent with tool execution on web chat and WhatsApp	Tested
RAG	Knowledge base retrieval for ONDC taxonomy and TEAM scheme FAQs	Tested
Multi-Channel	Voice → WhatsApp handoff for missing field collection	Tested

PoC Demo Note: For the PoC demonstration, voice interaction is implemented via browser-based Web Speech API with Sarvam AI STT/TTS due to infrastructure constraints. In production, the primary channel will be **telephony-based voice calls via Twilio/Exotel integration**, enabling MSE owners to add products by simply calling a toll-free number from any phone **without internet or smartphone requirement**.

4. Planned Pilot Deployment (Stage 1)

Target: Post Stage 1 Selection (INR 25 Lakhs)

Parameter	Planned Configuration
Region	Pan-India (MSE sellers across all states registered under TEAM scheme)
Infrastructure	NIC Cloud / MeghRaj Government Cloud Infrastructure
Primary Channel	Toll-free voice calls via Twilio/Exotel telephony (no internet required)
Secondary Channel	WhatsApp for catalogue completion, missing field collection, status updates
Tertiary Channel	Web dashboard for MSE profile management, SNP selection, admin verification
Target Users	MSEs registered under TEAM scheme seeking ONDC onboarding via SNPs
Integration Point	TEAM portal (team.msmemart.com) for MSE registration data validation

Pilot Scope:

- **500 MSEs** onboarded across 5 pilot states (Maharashtra, Gujarat, Rajasthan, UP, Tamil Nadu)
- **6 SNP integrations** — GlobalLinker, Plotch, SellerApp, eSamudaay, DotPe, MyStore
- **10 ONDC categories** covered (RET10–RET19) with full attribute validation
- **Hindi + English + Hinglish** voice support with Sarvam AI STT/TTS
- **WhatsApp Business API** for catalogue follow-up and milestone notifications
- **NSIC admin dashboard** for AI-assisted verification queue and funnel analytics

Pilot Success Metrics:

- Product classification accuracy > 90% across ONDC taxonomy
- Average onboarding time reduced from 14 days (manual) to < 3 days
- Catalogue compliance score > 85% before SNP handoff
- MSE-to-SNP match acceptance rate > 70%
- Voice channel adoption > 40% among rural MSEs

5. Planned Scale Deployment (Stage 2)

Target: Post Challenge Win (INR 1 Crore Contract - 2 Years)

Upon winning the IndiaAI Innovation Challenge, VyapaarSetu will be scaled for national deployment with the following enhancements:

- **Full TEAM Portal Integration:** Direct integration with NSIC's TEAM portal (team.msmemart.com) for real-time MSE registration sync, Udyam verification via live API, and automated SNP allocation
- **ONDC Network Connectivity:** Direct API integration with ONDC Gateway for real-time catalogue push to SNP seller nodes, eliminating manual catalogue re-entry
- **Regional Language Expansion:** Beyond Hindi/English to include Tamil, Telugu, Bengali, Marathi, Gujarati, and other scheduled languages via Sarvam AI
- **IVR-Based Voice Menu:** Interactive Voice Response system for product category selection before AI agent handoff, enabling feature phone users
- **Load-Balanced Telephony:** Scalable telephony infrastructure handling concurrent voice calls across India

Planned Production Architecture:

Layer	Components	Technology
Telephony Gateway	Toll-free number, IVR menu, call routing	Twilio / Exotel
Load Balancer	Traffic distribution, SSL termination	NIC Cloud LB / HAProxy
Application Tier	FastAPI instances (auto-scaled)	Kubernetes + Docker
AI/ML Services	LangChain agents, Sarvam AI STT/TTS, DeepSeek LLM	GPU-enabled pods
Messaging	WhatsApp Business API	Meta Cloud API
Data Tier	PostgreSQL (primary-replica), Redis Cluster	Managed databases
Vector Store	Qdrant cluster for RAG	Distributed Qdrant
Monitoring	Onboarding analytics, logging, metrics, alerting	Prometheus + Grafana

Scale Targets (2-Year Contract):

- **5,00,000 MSEs** onboarded onto ONDC (aligned with TEAM scheme target)
- **50% women-led MSEs** prioritised (TEAM scheme mandate)
- **All 36 states/UTs** covered with regional language support

- **20+ SNP integrations** across all ONDC retail categories
- **< 48 hours** average end-to-end onboarding time (registration to ONDC live)